



RISKS-X

风险分析预测与管控研究院

Unifying the Gutenberg-Richter Law with Probabilistic Catalog Completeness

Jiawei Li*, Xinyi Wang*(Presenter), and Didier Sornette

* Equal contribution

Institute of Risk Analysis, Prediction and Management (Risks-X), SUSTech
School of Computer Science, CUIT

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Motivation

Core Challenges in FMD Modeling

GR Law Limitations:

- Empirical FMD deviates from GR at low magnitudes ($m < M_c$)
- Fixed m_c approach oversimplifies gradual detection probability
- Deviations from the GR law are more likely observational artifacts than intrinsic physical effects.

Critical Need:

- Unified framework coupling detection probability & GR distribution
- Resolve asymmetry, nonlinearity, parameter interdependence

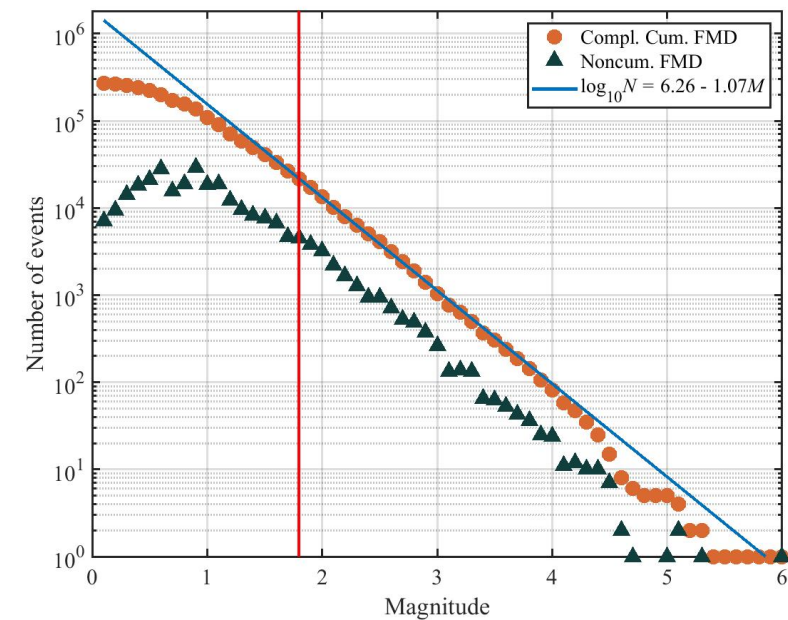


Figure 1. Complementary cumulative distribution (CCFMD) and density distribution (FMD) alongside the Gutenberg-Richter (GR) law, fitted using simulated earthquakes with magnitudes larger than 1.8.

MODELS

An augmented Gutenberg-Richter (GR) law tailored to handle incompleteness, expressed as:

$$\text{CCFMD}(m) = 10^{a-b \cdot G(m)} \quad (1)$$

in which,

$$G(m) = m_c + \sigma_c \cdot \text{gReLU}(x) \quad (2)$$

where $x = (m - m_c)/\sigma_c$.

GR law:

$$\text{CCFMD}(m) = 10^{a - bm} \quad (3)$$

1. m_c : 将完整性震级替换为地震检测快速变化的过渡震级
2. σ_c : 量化发生此转换的幅度范围。

gReLU

SSReLU.

$$\text{SSReLU}(x) = \ln(1 + e^x) \quad (4)$$

BSReLU.

$$\text{BSReLU}(x) = \left(x + \frac{m_c}{\sigma_c} \right) \Phi \left(\frac{\ln [(x\sigma_c + m_c) / m_c]}{\sigma_c} + \frac{\sigma_c}{2} \right) - \frac{m_c}{\sigma_c} \Phi \left(\frac{\ln [(x\sigma_c + m_c) / m_c]}{\sigma_c} - \frac{\sigma_c}{2} \right) \quad (5)$$

CORReLU.

$$\text{CORReLU}(x) = \begin{cases} e^{x-1} & x < 1 \\ x & x \geq 1 \end{cases} \quad (6)$$

AERReLU.

$$\text{AERReLU}(x) = \begin{cases} \frac{e^x}{1+\beta} & x < 0 \\ x + \frac{e^{-\beta x}}{1+\beta} & x \geq 0 \end{cases} \quad (7)$$

Performance of different models

gReLU — generalized Rectified Linear Unit

$$\text{ReLU}(x) = \max(0, x)$$

Here, we aim to seek functions that provide a smooth transition between the GR law for $m \geq m_c$ and a gradual decrease in recorded earthquake frequencies for $m < m_c$.

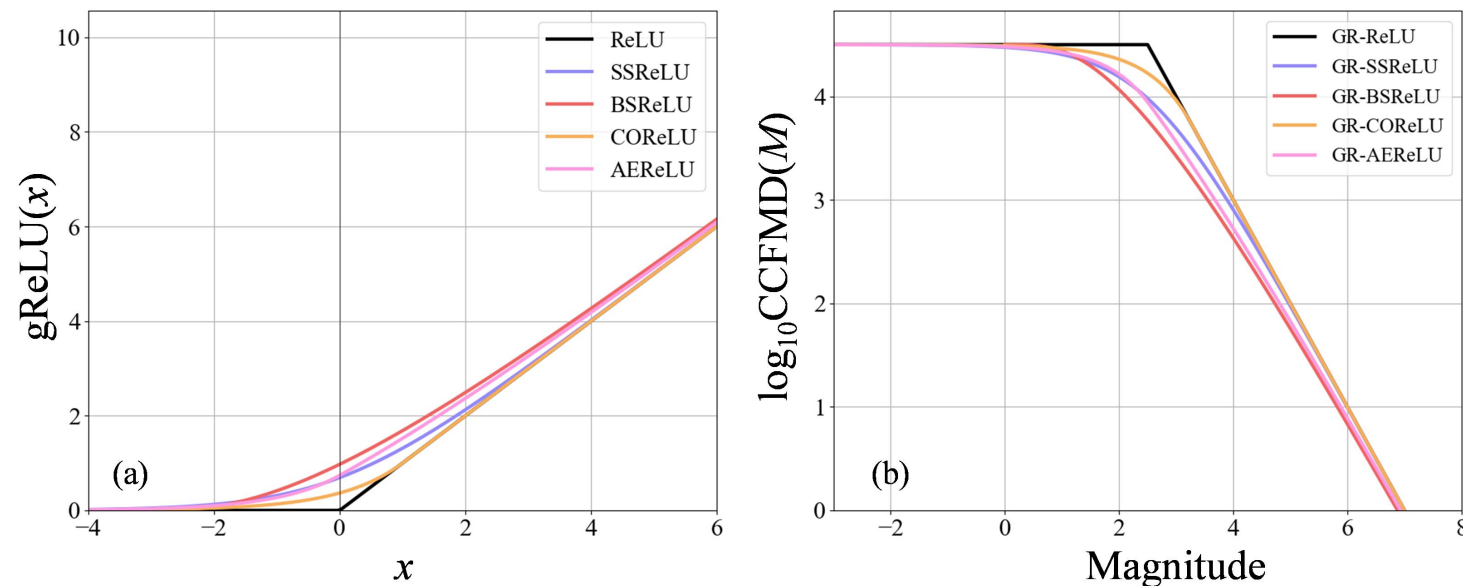
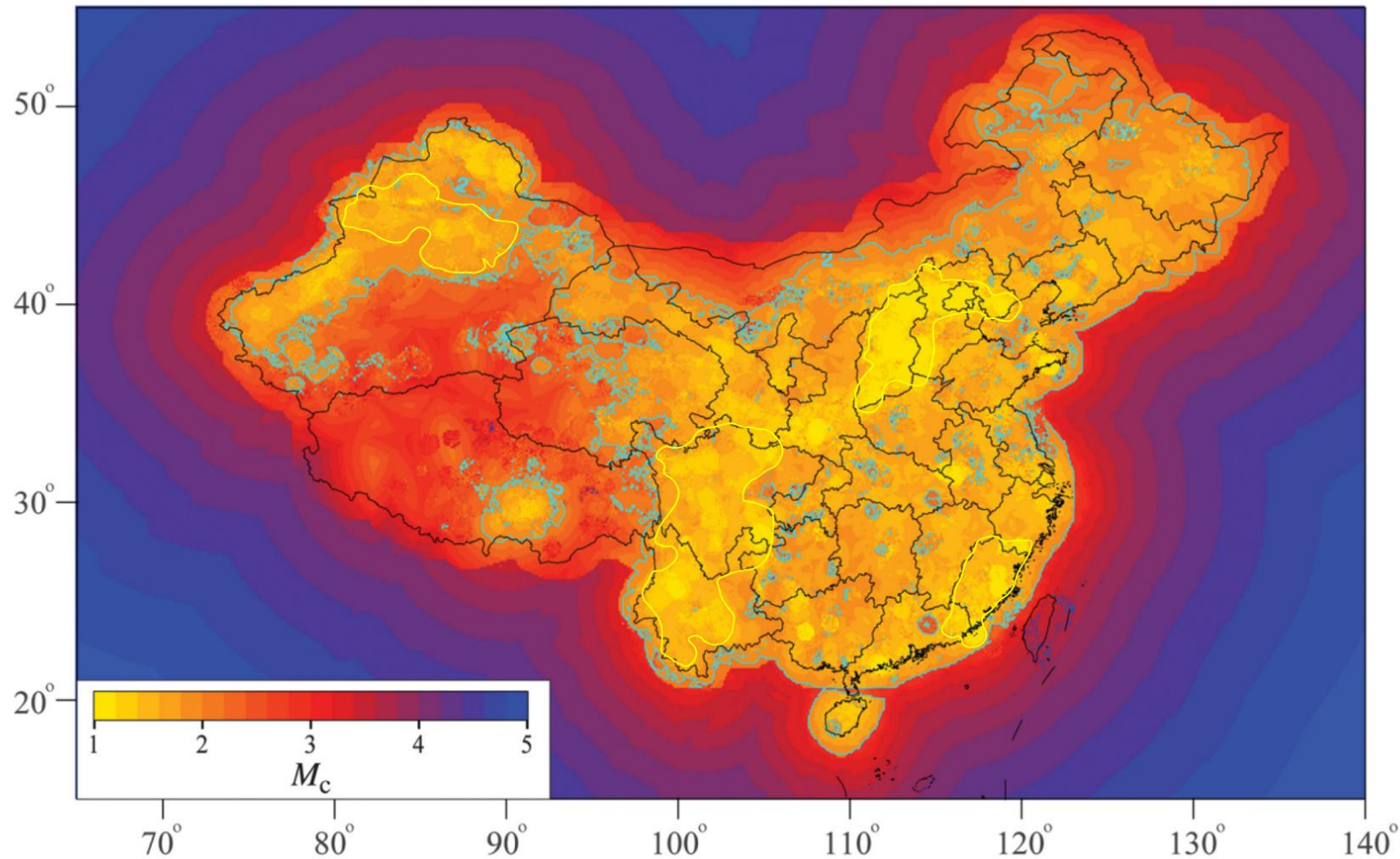


Figure 2. Plots of (a) the five gReLU(x) functions and (b) the corresponding augmented CCFMD. The parameters are set to $a = 7.0$, $b = 1.0$, $m_c = 2.5$, $\sigma_c = 0.75$, and $\beta = 0.35$.

Data: Synthetic & Observed Catalogs

Synthetic Catalog

- Generation:
Sampled from:
PDF bounds
Fixed param
- Design:
16 datasets($\geq$
200 realizations)



Results

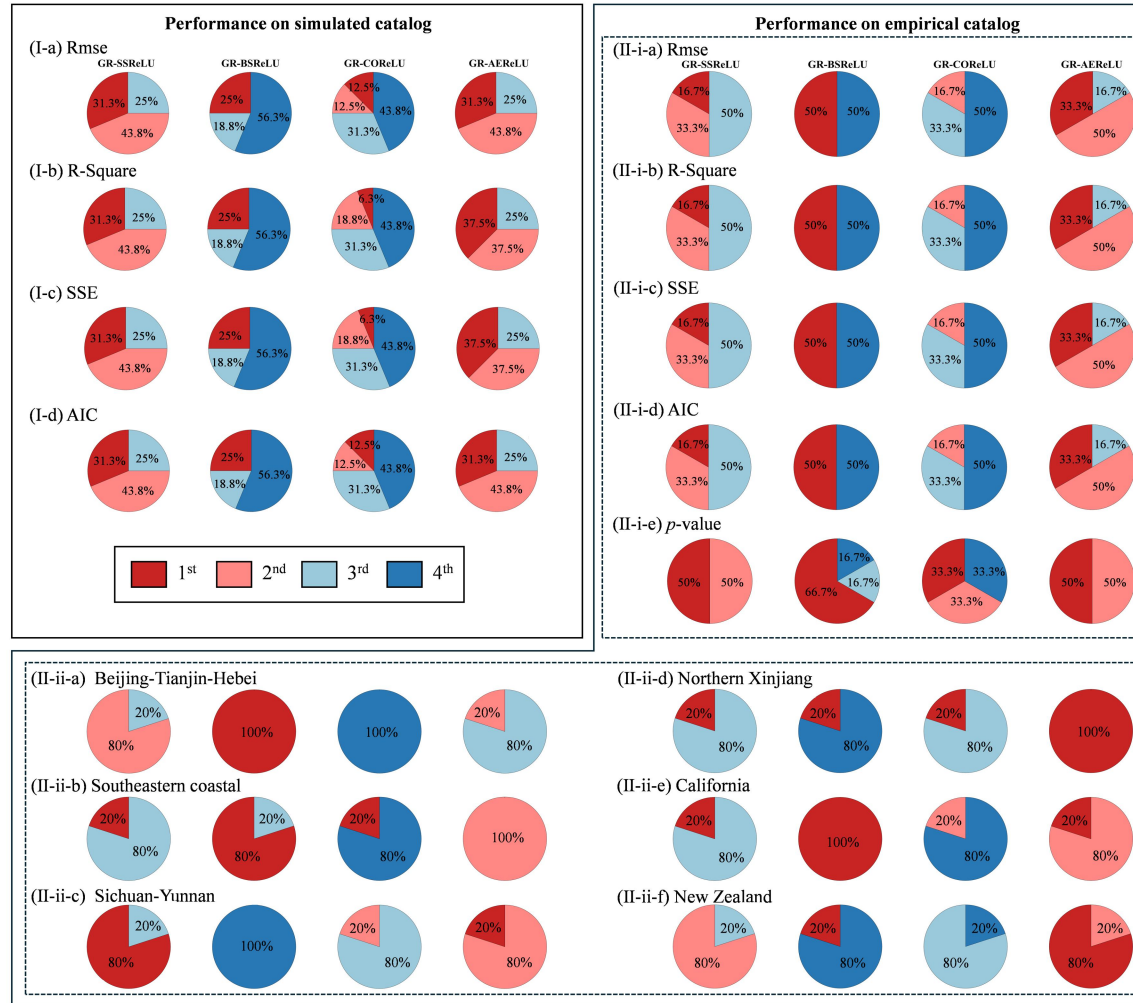


Figure3. The proportion of times each model ranked as best (1st, dark red), good (2nd, light red), poor (3rd, light blue), or worst (4th, dark blue) for simulated and empirical catalogs.

$$\ln[\text{PDF}_{\text{weight}}(b^{\text{pred}})] = \frac{1}{4} \sum_{i=1}^4 \omega_i \ln[\text{PDF}_i(b^{\text{pred}})]$$

$$\omega_i = \frac{\exp(-\frac{1}{2} \Delta \text{AIC}_i)}{\sum_{j=1}^4 \exp(-\frac{1}{2} \Delta \text{AIC}_j)}$$

with $\Delta \text{AIC}_i = [\text{AIC}_i - \min(\text{AIC})] / [\max(\text{AIC}) - \min(\text{AIC})]$

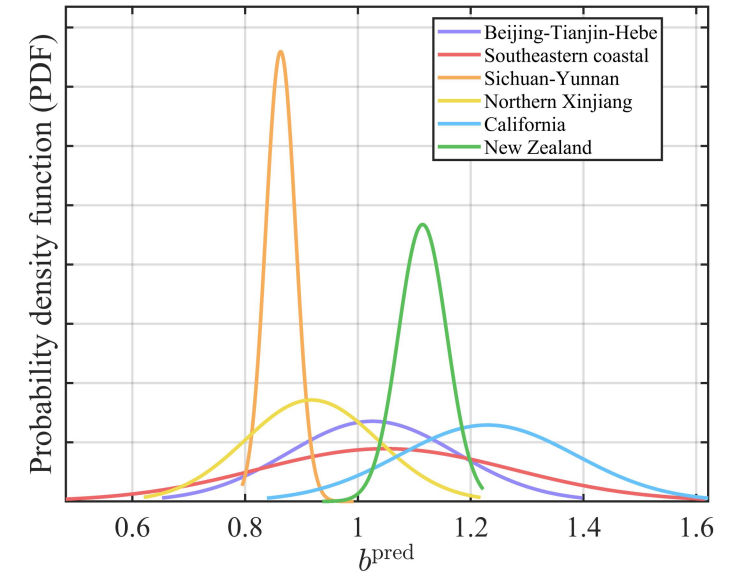


Figure4. The weighted averages of the b^{pred} -value PDFs from the four augmented GR models for the six study regions.

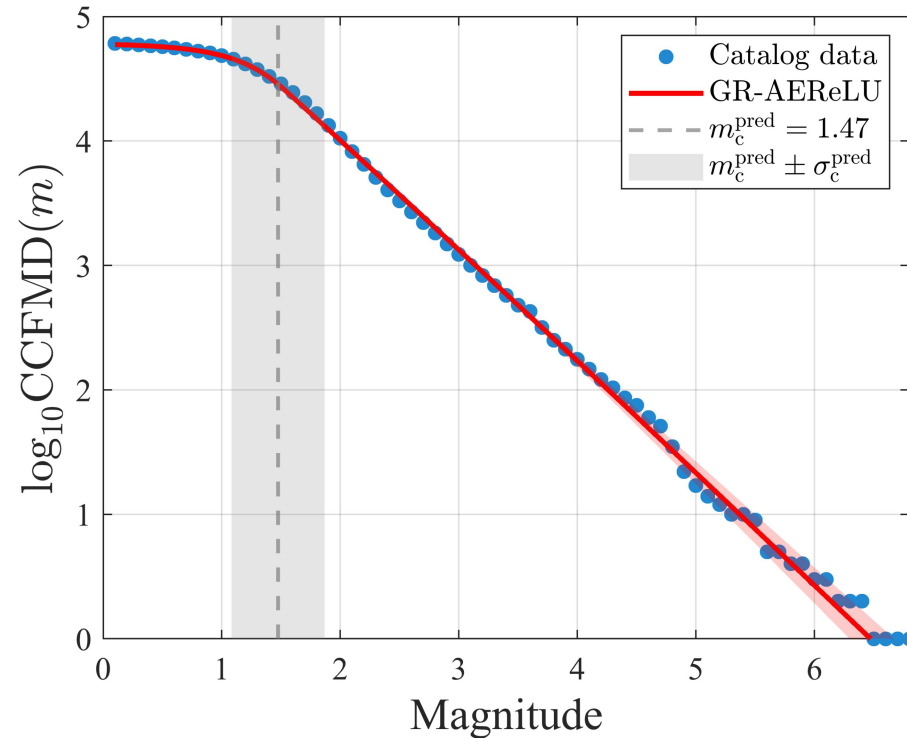


Figure 5. CCFMD of the empirical earthquake catalogs from northern Xinjiang, along with the fitted CCFMD(m) derived using the augmented Gutenberg-Richter law with the AEReLU(x) function. The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

Conclusion

The catalog completeness probability $P(m)$: the probability that events of magnitude m or larger are recorded in a given region.

$$P(m) = \frac{10^{a-b \cdot G(m)}}{10^{a-b \cdot m}} = 10^{b[m-G(m)]}$$

Substituting the expression for $G(m)$, we obtain:

$$P(m) = 10^{b\sigma_c \left[\frac{m-m_c}{\sigma_c} - \text{gReLU}\left(\frac{m-m_c}{\sigma_c}\right) \right]}$$

Thus, by substituting the $\text{gReLU}(x)$ expressions and the corresponding parameter estimates into Equation, one can compute the completeness probability for any given magnitude m .

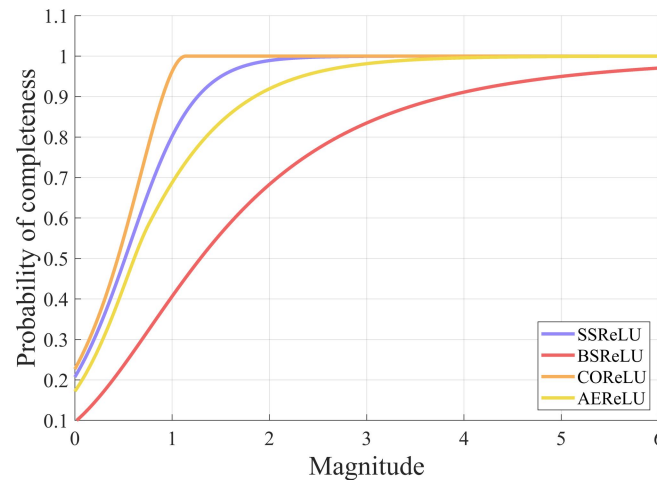


Figure 5. Probability of completeness $P(m)$ for the of the Beijing-Tianjin-Hebei region.

Results

Estimating M_c for simulated seismicity with homogeneous/heterogeneous M_c

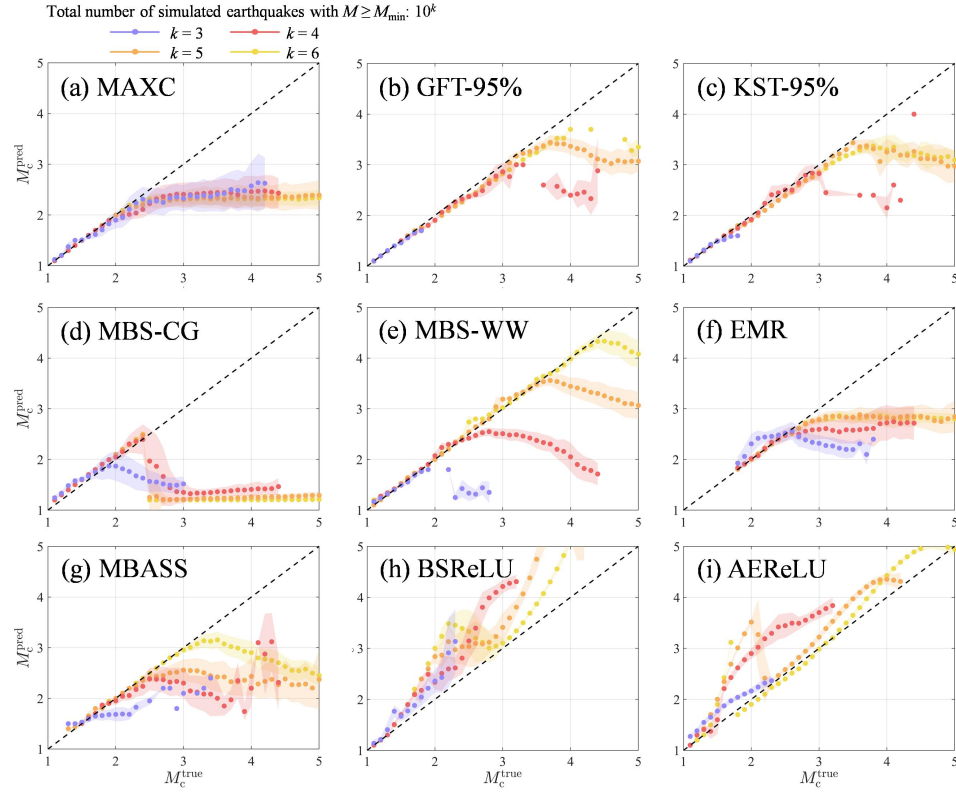


Figure 5. Results of the estimation of the completeness magnitude M_c^{pred} as a function of the true completeness magnitude M_c^{true} for the simulated datasets with homogeneous M_c , after being iteratively pruned three times. The shaded areas represent the uncertainties in the estimated completeness magnitude M_c^{pred} . In panels (h) BSReLU and (i) AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%. The different colors correspond to the total number of simulated earthquakes with $M \geq M_{\min} = 1$, with 10^k for $k = 3, 4, 5, 6$.

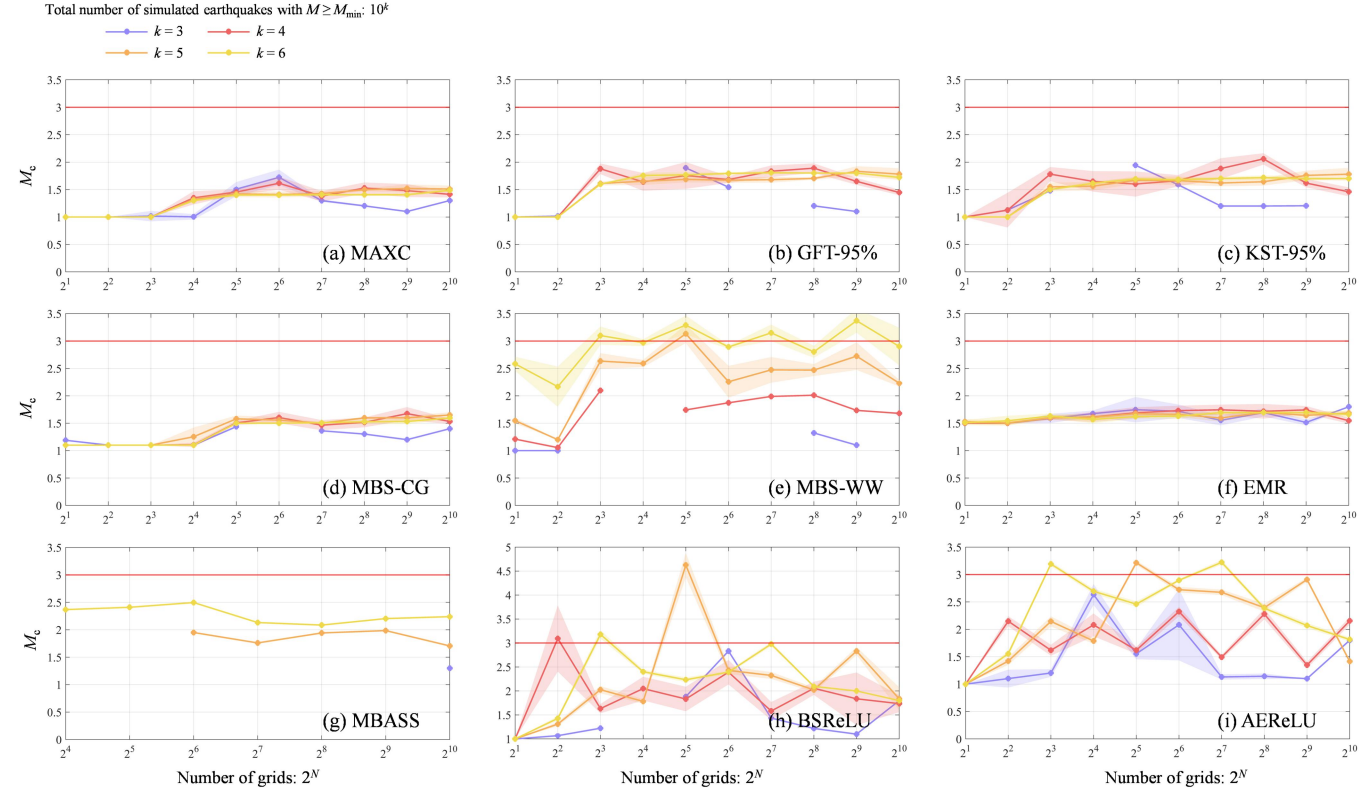


Figure 6. Results of the estimation of the completeness magnitude M_c as a function of the number of grids used to distribute the simulated earthquakes, after being iteratively pruned three times. The total number of simulated earthquakes with $M \geq M_{\min}$, given by 10^k for $k = 3, 4, 5, 6$, is randomly and uniformly distributed across 2^N ($N = 1, 2, \dots, 10$) grids. The minimum completeness magnitude ($M_{\min} = 1.0$) and maximum completeness magnitude ($M_c^{\max} = 5.0$) are assigned to the first and last grids, respectively, with the remaining grids having completeness magnitudes linearly interpolated between these values. For example, with $2^N = 2^2$ grids, the completeness magnitudes are $M_{c,\text{grid}1} = 1.0$, $M_{c,\text{grid}2} = 2.3$, $M_{c,\text{grid}3} = 3.7$, and $M_{c,\text{grid}4} = 5.0$, ensuring that the average M_c across all grids is 3.0 (indicated by the red horizontal line in the figure). The shaded areas represent the uncertainties in the estimated completeness magnitude. The different colors correspond to the total number of simulated earthquakes.

Key Points

- (1) We propose an augmented modeling framework that unifies the Gutenberg-Richter law with a probabilistic treatment of catalog incompleteness.
- (2) We evaluate four variants of the proposed Gutenberg-Richter model using both synthetic and empirical catalogs across multiple performance metrics.
- (3) The GR-AEReLU, allowing for an asymmetric crossover with two distinct exponential decays, consistently outperforms other variants and offers valuable seismological insights.
- (4) The earthquake b -value varies systematically by region and tectonic setting, not the universal constant of 1.0 assumed in standard seismology.

THANKS

presenter: Xinyi Wang
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